

fit@hcmus

Software Testing

CSC13003

Performance Testing

Content

- What is Performance Testing?
- Why to do Performance Testing?
- Types of Performance Testing
- How to do Performance Testing?

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- **What is Performance Testing?**
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What is Performance Testing?

Performance testing is a type of **non-functional testing** that ensures software applications to **perform properly** under their **expected workload**.

What is Performance Testing?

- The main goal is
 - not to find bugs
 - to eliminate performance bottlenecks
- The focus is on
 - **speed** – whether the application responds quickly
 - **scalability** – the maximum user load the application can handle
 - **stability** – if the application is stable under varying loads

What is Performance Testing?

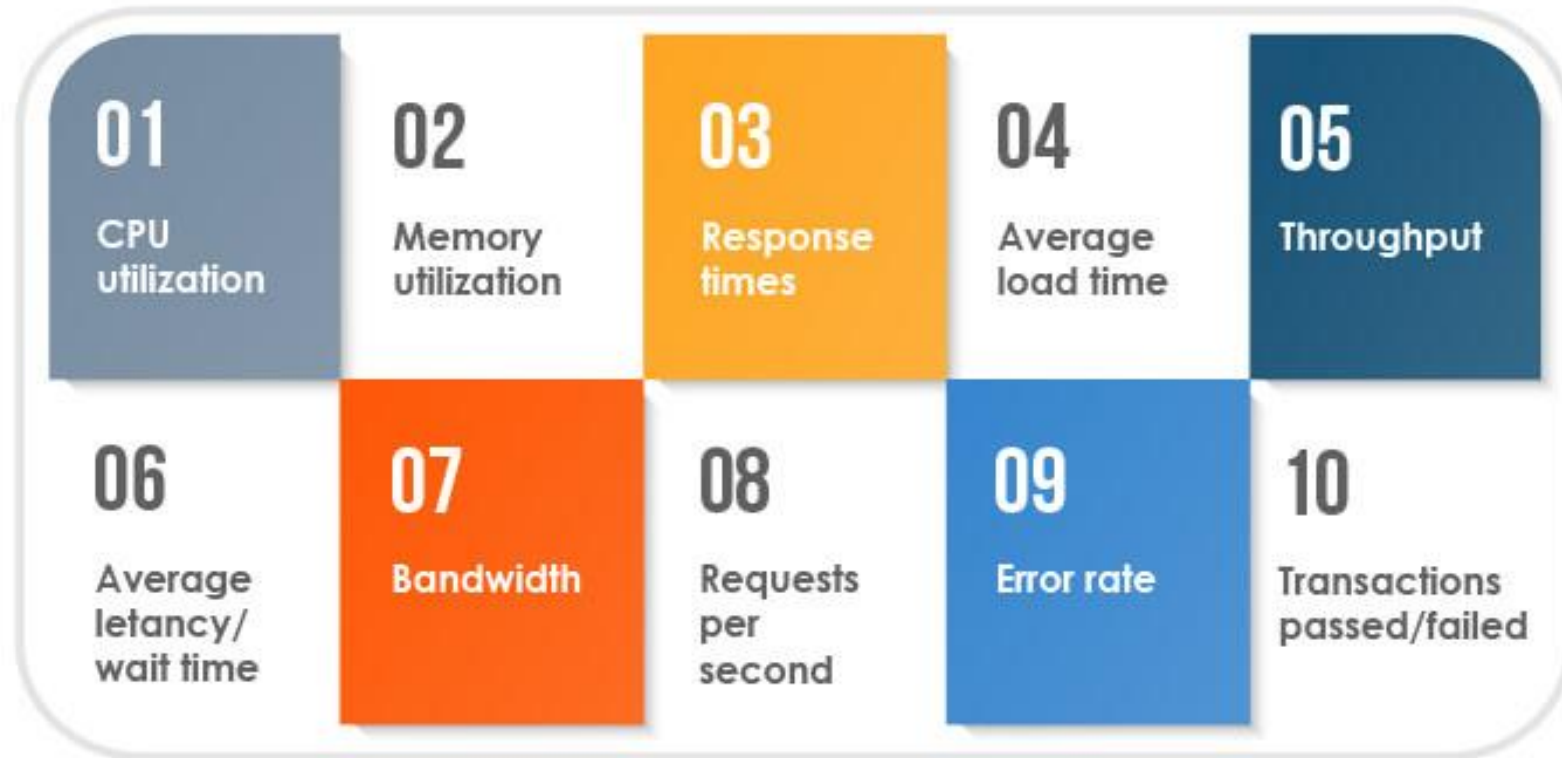
- Common Performance Problems
 - Long load time – long initial time to start an application
 - Poor response time – delayed output response to an input
 - Poor scalability – does not support a large enough number of users
 - Bottlenecks – obstacles that degrade overall system performance

What is Performance Testing?

- Example Performance Test Cases
 - Verify response time is not more than 4 secs when 1000 users access the website simultaneously
 - Verify response time of the Application Under Load is within an acceptable range when the network connectivity is slow
 - Check the maximum number of users that the application can handle before it crashes
 - Check database execution time when 500 records are read/written simultaneously
 - Check CPU and memory usage of the application and the database server under peak load conditions
 - Verify the response time of the application under low, normal, moderate, and heavy load conditions

What is Performance Testing?

- Performance Testing Metrics



What is Performance Testing?

- Performance Testing Metrics
 - CPU utilization – percentage of CPU capacity utilized
 - Memory utilization – utilization of the primary memory
 - Response times – time between sending request and receiving response
 - Average load time – time to complete the loading process
 - Throughput – the number of transactions can be handled in a second

What is Performance Testing?

- Performance Testing Metrics

- **Average latency/Wait time** – the time spent by a request in a queue before getting processed
- **Bandwidth** – the volume of data transferred per second
- **Requests per second** – the number of requests handled per second
- **Error rate** – the percentage of requests resulting in errors
- **Transactions Passed/Failed** – the percentage of passed/failed transactions

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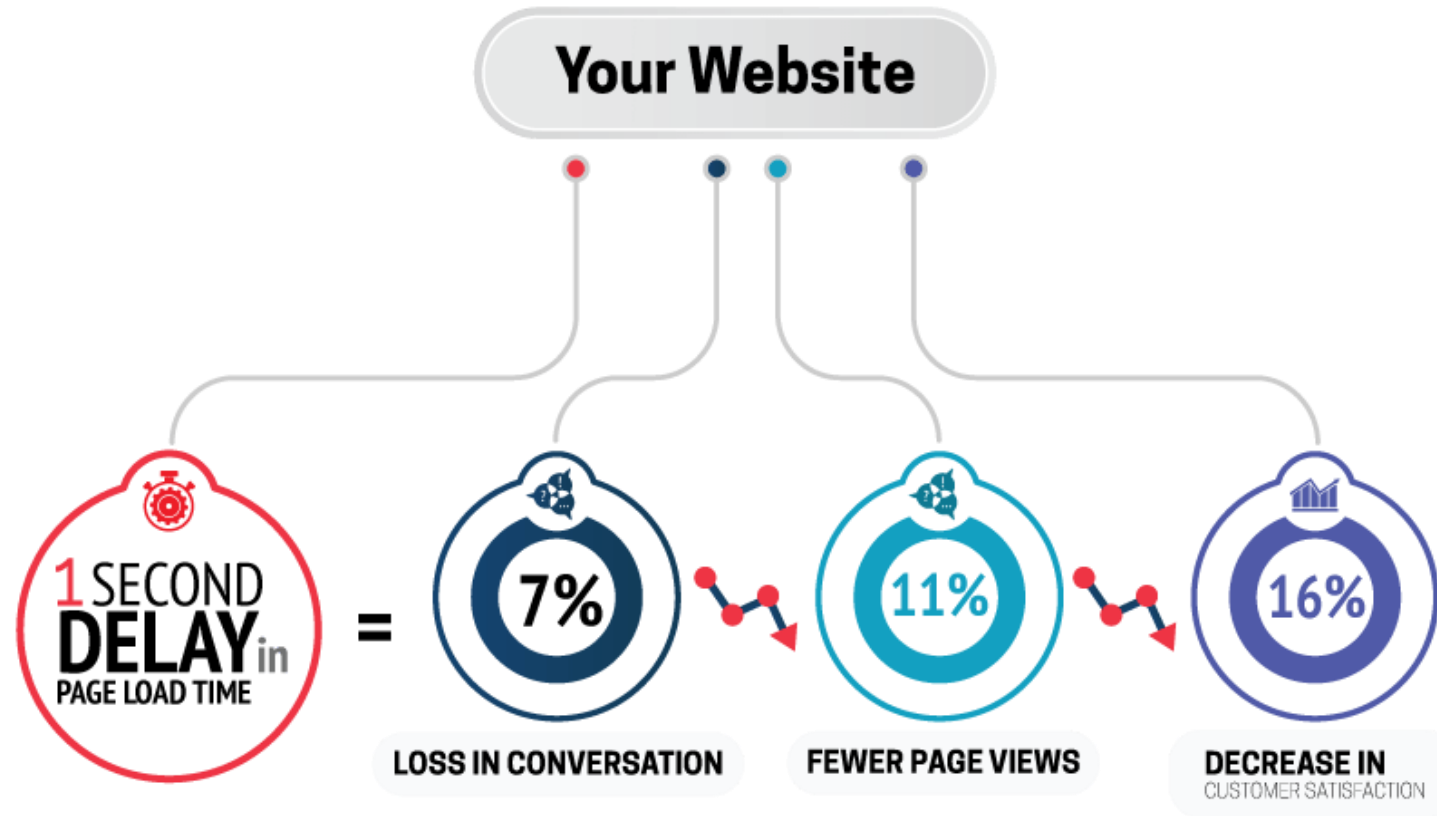
Why to do Performance Testing?

- Most users click away after **8 seconds of delay**
- **\$4.4 billion business revenue loss** due to poor web applications performance
- Aberdeen found that inadequate performance could **impact revenue by up to 9%**
- Business performance begins to suffer at **5.1 seconds of delay in response times** of web applications and 3.9 for critical applications

Why to do Performance Testing?

Only a **5-minute downtime** of Google.com (19-Aug-13) is estimated to cost the search giant as much as **\$545,000**.

It's estimated that companies lost sales worth **\$1100 per second** due to a recent Amazon Web Service Outage.



IN DOLLAR TERMS,
this means that if your site typically earns \$100,000 a day,
this year you could lose **\$2.5 MILLIONS** in sales.

Why to do Performance Testing?

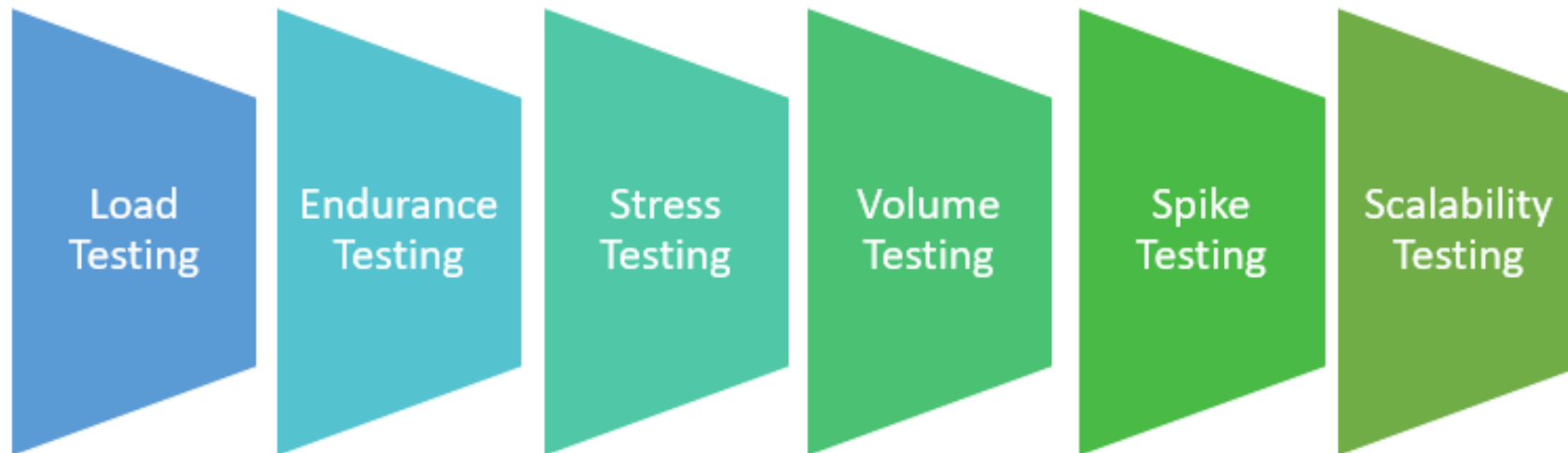
- Help ensure the software
 - meet the expected levels of service
 - provide a positive user experience
- Highlight improvements relative to speed, stability, and scalability
- Absence of testing might suffer from different types of problems that lead to a damaged brand reputation

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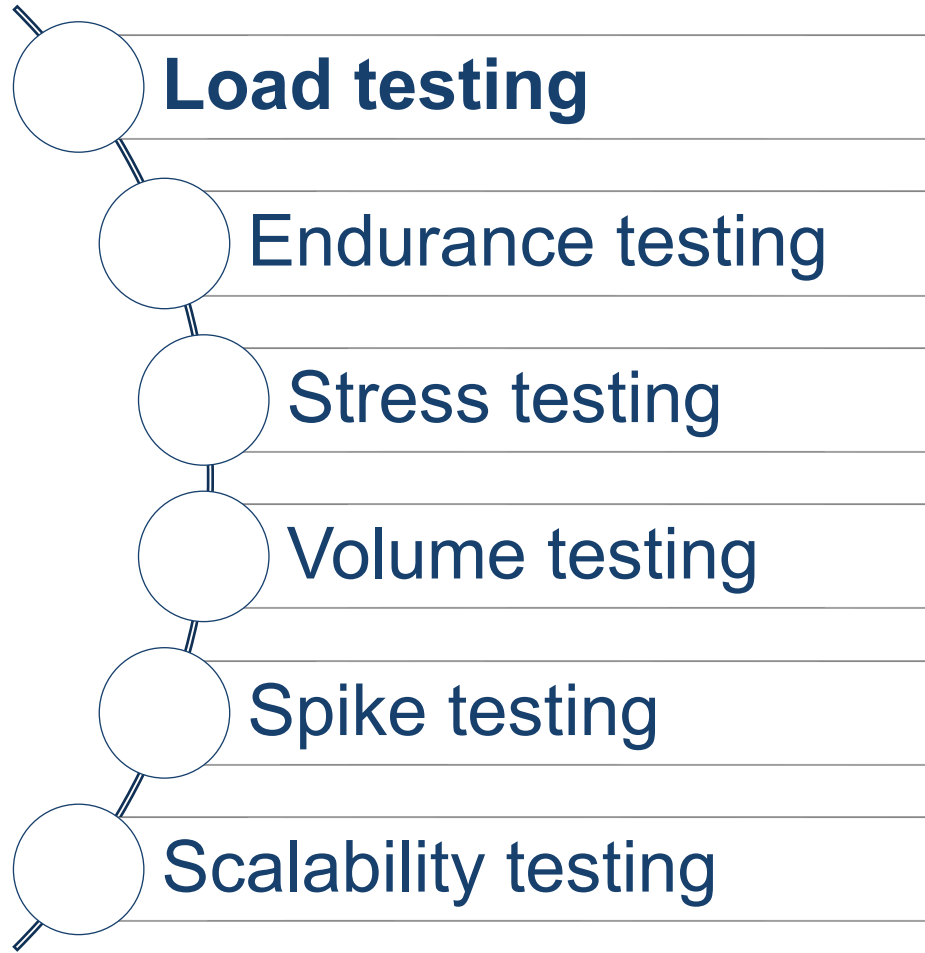
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Types of Performance Testing

Types of Performance Testing

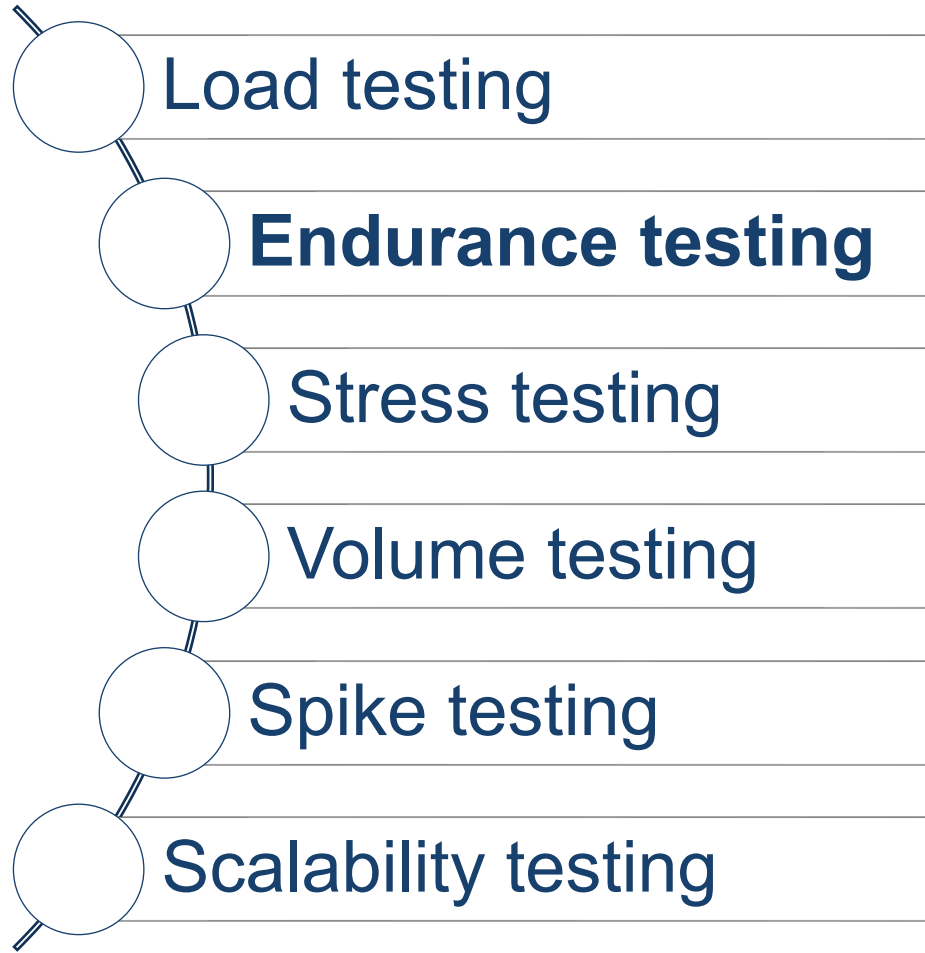


Types of Performance Testing



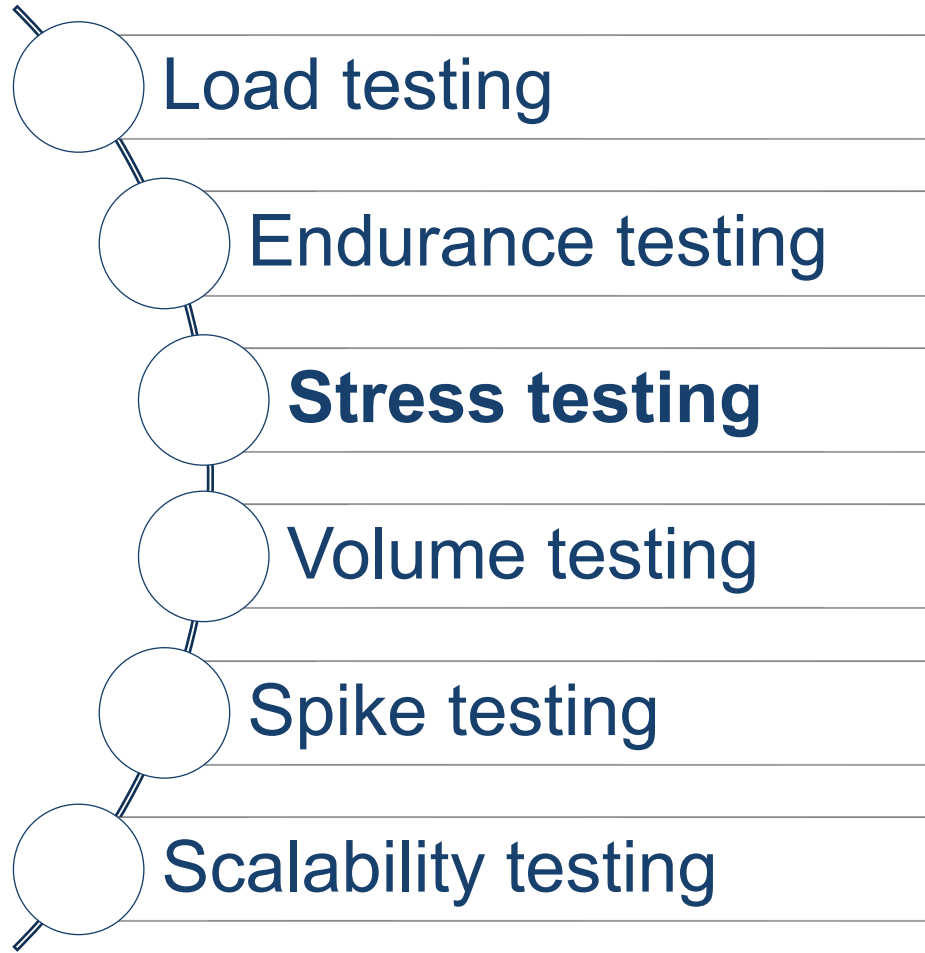
- It checks the product's ability to perform under anticipated user loads
- The objective is to identify performance congestion before the software product is launched in market

Types of Performance Testing



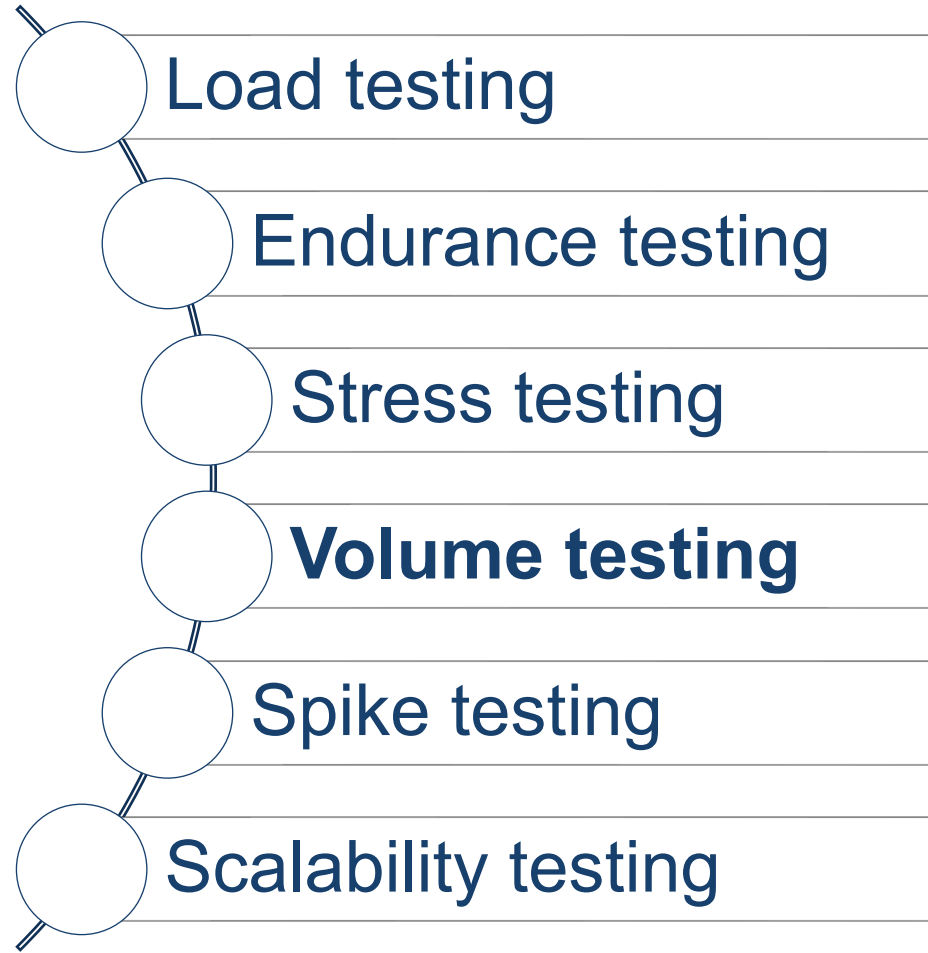
- It is performed to ensure the software can handle the expected load over a long period of time

Types of Performance Testing



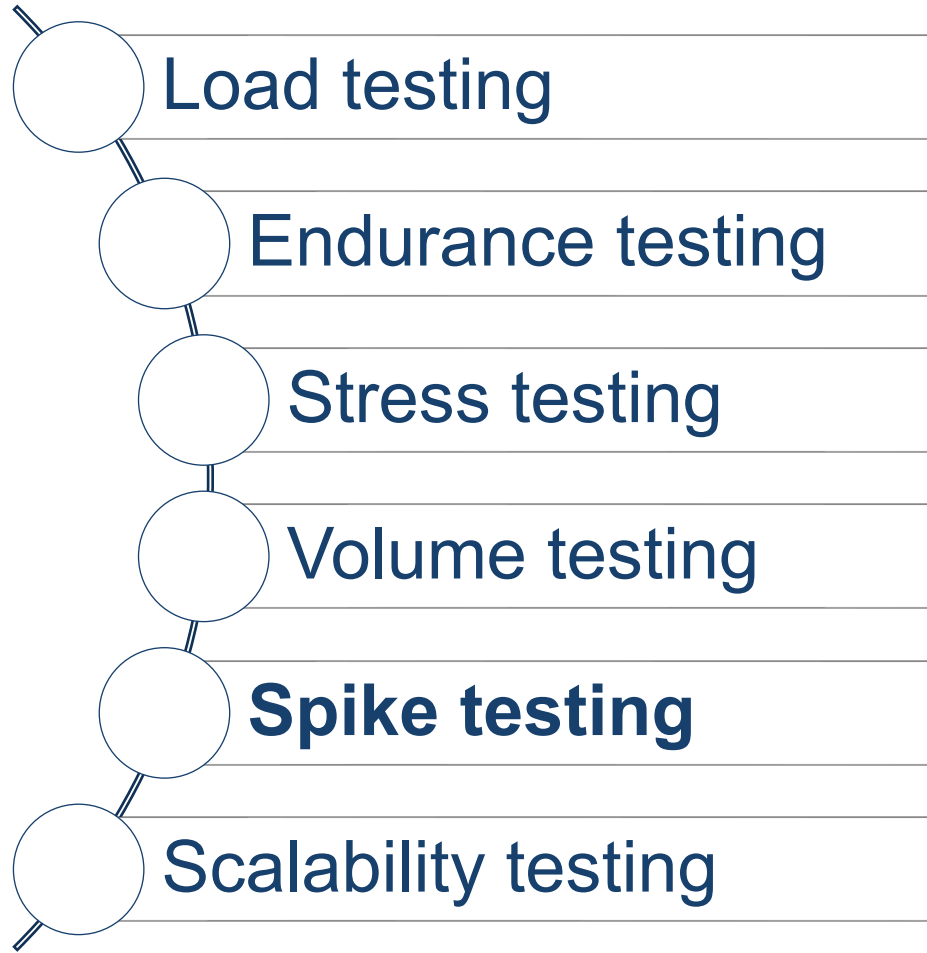
- It involves testing a product under extreme workloads to see whether it handles high traffic or not
- The objective is to identify the breaking point of a software product

Types of Performance Testing



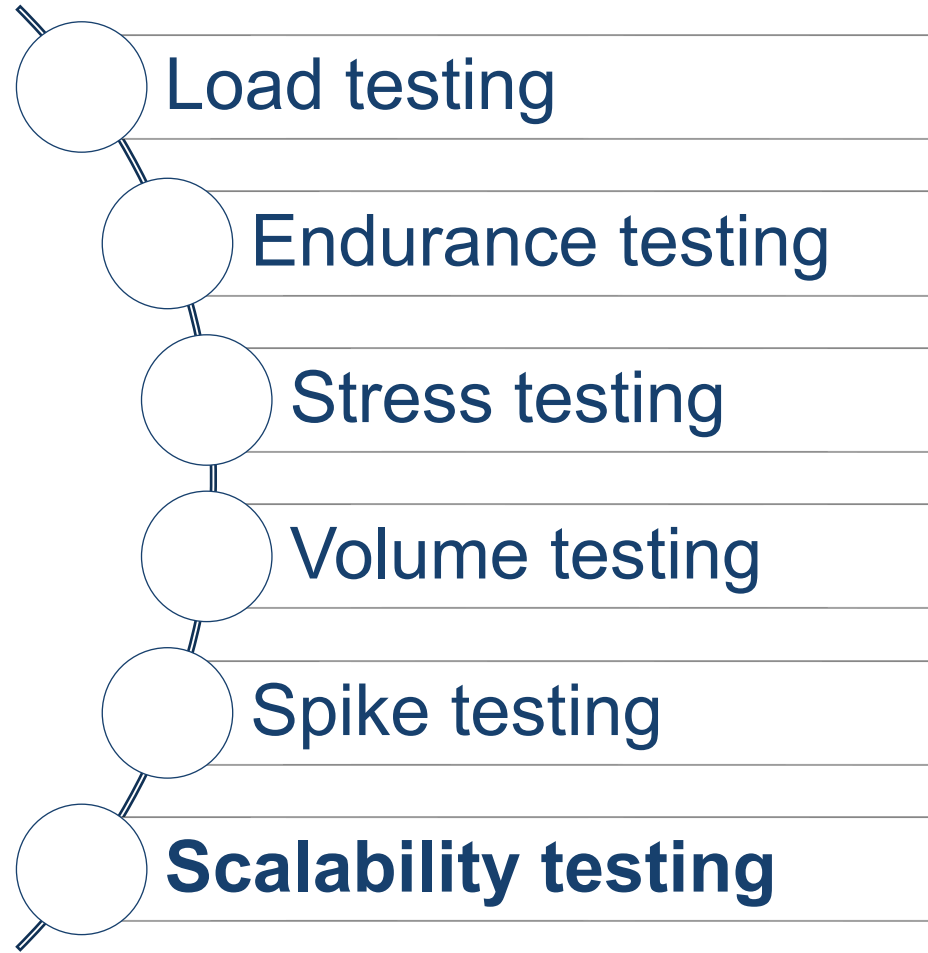
- In volume testing large number of data is saved in a database and the overall software system's behavior is observed
- The objective is to check product's performance under varying database volumes

Types of Performance Testing



- It tests the product's reaction to sudden large spikes in the load generated by users

Types of Performance Testing

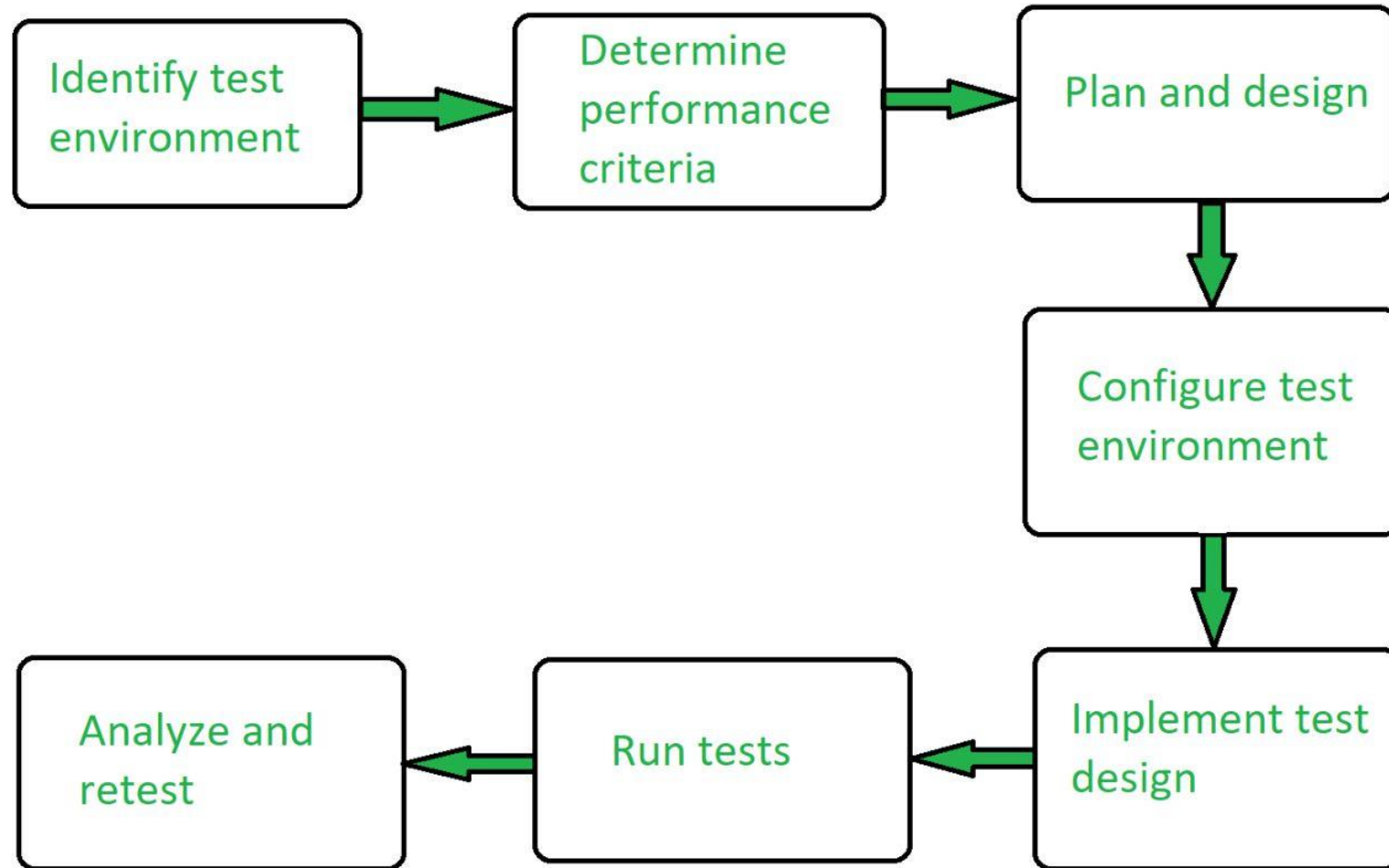


- In scalability testing, software application's effectiveness is determined in scaling up to support an increase in user load
- It helps in planning capacity addition to your software system

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How to do Performance Testing?



How to do Performance Testing?



Step 1 – Identify test environment

- Identify the testing environment and know what testing tools are available at your disposal
- Understand the details of all the hardware, software and different network configurations ahead of time

How to do Performance Testing?



Step 2 – Determine performance criteria

- Identify the general performance metrics
- Identify the performance success criteria

How to do Performance Testing?



Step 3 – Plan and design

- Identify key scenarios by considering
 - user variability
 - test data
 - plan performance
- Simulate a variety of use cases
- Outline what metrics will be gathered

How to do Performance Testing?



Step 4 – Configure test environment

- Arrange all the necessary testing tools and monitoring resources

Step 5 – Implement test design

- Design performance tests according to performance criteria and metrics

How to do Performance Testing?



Step 6 – Run tests

- Execute and monitor the performance tests

Step 7- Analyze and retest

- Analyze the finding and fine tune the test again to see an increase or decrease in performance
- Run the tests again using the same or different parameters

